

Molecular Beam Epitaxy of AgTaO₃

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We report the first synthesis of single-crystal AgTaO₃ thin films using molecular-beam epitaxy (MBE). High-quality epitaxial AgTaO₃ films were grown on both (001)- and (111)-oriented SrTiO₃ substrates using sequential deposition of atomic silver and TaO₂ layers under an ozone/oxygen atmosphere (80 % O₃ + 20 % O₂). X-ray diffraction and reciprocal space mapping demonstrate that the films are coherently strained to the SrTiO₃ substrates with sharp rocking curves comparable to the substrates, indicating high structural perfection. High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) confirms coherent, low defect growth for (001)_{pc}-oriented films. For (111)_{pc}-oriented films, initial coherent growth proceeds up to 10 nm before transitioning into a Ta-rich surface region. Energy-dispersive X-ray spectroscopy (EDX) reveals a narrow cation intermixing region at the substrate interface for both orientations. This work demonstrates an effective synthesis route for single-crystal AgTaO₃ thin films, providing a platform to investigate strain engineering and emergent interfacial properties in silver-based tantalates.

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I. INTRODUCTION

Tantalate perovskites have emerged as a promising material platform to explore quantum phenomena in bulk and at oxide interfaces, driven by the combination of strong 5d spin-orbit coupling and enhanced permittivity and carrier mobility.^{1–3} When integrated into low-dimensional heterostructures, these features could give rise to two-dimensional electron gases^{4,5} with robust interfacial superconductivity^{6–8} and efficient spin-charge interconversion.^{9,10} Realizing the full potential of these emergent properties, however, hinges on the ability to synthesize single-crystal thin films with precise stoichiometry, atomic-scale interface sharpness, and minimal defect densities.¹¹ Despite recent interest in quantum phases in KTaO₃,^{12–14} less attention has been given to the Ag⁺ analog. AgTaO₃ exhibits relatively high permittivity, low loss tangent and potential as an anti-ferroelectric storage material^{15–17}.

First synthesized by Francombe and Lewis in 1958,¹⁸ the room-temperature crystal structure of AgTaO₃ has been a subject of debate. Initially, the room temperature structure was believed to be orthorhombic.¹⁸ Later, Belyaev assigned a polar rhombohedral structure (*R3c*) to room temperature AgTaO₃.¹⁹ A permittivity anomaly typically marks the polar transition. Soon et al.²⁰ did not observe an anomaly in permittivity and proposed a centro-symmetric rhombohedral structure (*R $\bar{3}m$*) for room-temperature AgTaO₃, while others reported such anomaly.²¹ More recent diffraction results suggest the following sequence of phase transitions for bulk AgTaO₃: *R3c* $\xleftrightarrow{400^\circ\text{C}}$ *Ibmm* $\xleftrightarrow{430^\circ\text{C}}$ *P4/mbm* $\xleftrightarrow{500^\circ\text{C}}$ *Pm $\bar{3}m$* .²² Furthermore, epitaxial synthesis parameters such as epitaxial strain and substrate clamping effects could impact the structural phase diagrams in epitaxial thin layers.^{23–25}

In conventional solid-state synthesis, AgTaO₃ is formed by reacting Ag₂O and Ta₂O₅ powders. Thermo-gravimetric analysis performed by Valant et al. showed that Ag₂O decomposes to metallic silver before the perovskite formation, suggesting Ag⁺ is only recovered once silver reacts with Ta₂O₅ and oxygen at higher temperatures.¹⁶ Thin-film deposition techniques that supply individual atomic species could circumvent this thermodynamic barrier. Despite the successful bulk synthesis of AgTaO₃, its thin film synthesis has remained limited²⁶ and of this materials system by molecular-beam epitaxy (MBE) has not been demonstrated. A synthesis approach for high-quality single crystalline thin films could resolve competing structures²⁷ and provide a platform to investigate the impact of epitaxial tuning. Furthermore, recent molecular-beam epitaxy of close relative KTaO₃²⁸ has enabled the epitaxial

strain stabilization of a robust ferroelectric state above room temperature.²⁹

In this work, we present the first MBE synthesis of AgTaO₃. AgTaO₃ films are grown on SrTiO₃ (100) and SrTiO₃ (111) single crystal substrates. X-ray diffraction reveals that the films are coherently strained and have high quality, comparable to SrTiO₃ single crystal substrate. HAADF-STEM images confirm that the films have abrupt interfaces and do not show any extended defects.

II. EXPERIMENTAL

Epitaxial AgTaO₃ films were grown using a modified Veeco Gen 10 MBE system, where the conventional SiC heater was replaced by 10 μ m CO₂-laser from Epiray GmbH (THERMALAS Substrate heater). A molecular beam of TaO₂ was generated from an effusion cell containing Ta₂O₅ (Alfa Aesar, 99.993 %).^{30,31} A molecular beam of silver was generated by a conventional effusion cell (Alfa Aesar, 99.999 %). Films were grown by sequential deposition of one silver atomic layer followed by one atomic layer of TaO₂ in a substrate temperature range of 500-600 °C, measured by an optical pyrometer operating at 7.5 μ m wavelength. The silver flux was measured before growth using a quartz crystal microbalance ($1-2 \times 10^{13}$ atoms/cm²/s) while the TaO₂ was determined by deposition on R-plane sapphire. The deposition rate was confirmed by x-ray reflectometry measurements. A mixture of ozone and oxygen (80 % O₃ + 20 % O₂) was used as the oxidant. The films were grown at an oxidant background pressure of 1×10^{-5} Torr. The (001) and (111) SrTiO₃ substrates were used as received.

X-ray diffraction (XRD), X-ray reflectometry (XRR), and reciprocal space mapping (RSM) measurements were carried out using a PANalytical Empyrean diffractometer with Cu K α_1 radiation. The raw XRR spectra were analyzed using the PANalytical X'Pert Reflectivity software package and the layer thickness was derived from a fast Fourier transform (FFT) after manually defining the critical angle to account for refractive effects. *In situ* X-ray photoelectron spectroscopy (XPS) was performed in the same chamber using a non-monochromated Scienta Omicron DSX400 x-ray source. Spectra were analyzed with an Omicron Sphera II analyzer after excitation. *In situ* reflection high-energy electron diffraction (RHEED) patterns were recorded using KSA-400 software and a Staib electron source operated at 14 kV and a filament current of 1.5 A. The morphology of the film surface was

characterized using an Asylum Cypher ES environmental AFM. Cross-sectional TEM specimens were prepared by focused ion beam milling. Initial milling was performed at 30 kV, followed by final thinning and polishing at 5 kV to reduce ion-beam-induced surface damage. STEM imaging was carried out using a Thermo Fisher Scientific Themis Z operated at 200 kV with a probe semi-convergence angle of 20 mrad. High-angle annular dark-field STEM (HAADF-STEM) images were acquired over a detector collection range of 64–200 mrad. To improve the signal-to-noise ratio while minimizing the effects of sample drift, 20 fast-scan images, each containing 2048×2048 pixels with a dwell time of 200 ns per pixel, were aligned and averaged. Energy-dispersive X-ray spectroscopy (EDX) spectrum imaging was performed using a Super-X detector, and elemental maps were generated from the net X-ray counts.

III. RESULTS

The bulk AgTaO₃ lattice constants are $a = b = 5.528 \text{ \AA}$ and $c = 13.715 \text{ \AA}$ with rhombohedral space group symmetry ($R3c$, $a^-a^-a^-$ Glazer notation) at room temperature.³² Accordingly, AgTaO₃ grows pseudocube-on-cube on SrTiO₃. The expected crystallographic orientation of AgTaO₃ grown on SrTiO₃ (001) and (111) are (012)_R and (001)_R, receptively, corresponding to the (001)_{pc} and (111)_{pc}. Here the subscript R denotes rhombohedral indices and pc denotes pseudocubic indices. The lattice mismatch between (001)_{pc} AgTaO₃ ($a_{pc} = 3.9258 \text{ \AA}$) and (001) SrTiO₃ ($a_{STO} = 3.9051 \text{ \AA}$) is -0.53% . The lattice mismatch between (111)_{pc} AgTaO₃ ($a_{pc} = 5.5281 \text{ \AA}$) and (111) SrTiO₃ ($a_{STO} = 5.5226 \text{ \AA}$) is -0.1% . Reflection high-energy electron diffraction (RHEED) was used to monitor the evolution of the surface structure and reconstruction during growth of both epitaxial orientations. Figures 1(a) and 1(b) show the RHEED diffraction pattern after the growth of first formula-unit-thick of (001)_{pc} and (111)_{pc} AgTaO₃ films on (001) and (111) SrTiO₃, respectively. Along their high symmetry directions diffraction streaks and Kikuchi lines are visible. We note the small additional dots at (001)_{pc} AgTaO₃ orientation. Figures 1(c) and (d) show the RHEED pattern after the growth of a 20 formula-unit-thick layer of (001)_{pc} and (111)_{pc} AgTaO₃, respectively. The additional dot-like features observed in the (001)_{pc} orientation have vanished and clear diffraction pattern of AgTaO₃ is visible. Figures 1(e) and (f) exhibit the RHEED pattern immediately after the growth of 28 nm thick (001)_{pc} and

17 nm thick (111)_{pc} AgTaO₃ films, where the shutters of both silver and TaO₂ are closed, but the substrate is still immersed in ozone and at growth temperature. While the (001)_{pc} AgTaO₃ RHEED pattern remains clear, the (111)_{pc} AgTaO₃ appears cloudy.

In situ X-ray photoelectron spectroscopy (XPS) was used to investigate the oxidation states of the AgTaO₃ thin films (Fig. S1). The Ag 3*d* spectrum exhibits a spin-orbit doublet with the 3*d*_{5/2} and 3*d*_{3/2} components centered at 368.10 eV (FWHM = 1.58 eV) and 374.12 eV (FWHM = 1.55 eV), respectively, corresponding to a spin-orbit splitting of 6.01 eV and an intensity ratio of approximately 1.46:1 (close to the statistical 3:2 ratio). Assigning this binding energy to a specific silver oxidation state is complicated by substantial overlap reported in the literature.³³ The Ta 4*f* doublet at 4*f*_{7/2} = 26.25 eV and 4*f*_{5/2} = 28.12 eV is consistent with the Ta⁵⁺ oxidation state.^{2,34}

Atomic force microscopy (AFM) was used to investigate the surface *ex situ* (Figs. S2 and S3). The root-mean-square roughness, measured by taking a 1 μm² area as a reference, is 0.94 nm and 0.82 nm for the (001)_{pc} and (111)_{pc} oriented AgTaO₃ films, respectively. Figure 2 shows the x-ray diffraction results of the same 28 nm (001)_{pc} and 17 nm (111)_{pc} thick AgTaO₃ films. Figure 2(a) shows the θ -2 θ scan around 001 AgTaO₃ reflection, surrounded by Laue fringes.³⁵ The full spectra can be found in the supplemental material (Fig. S4 and S5). An RSM around the 103 substrate reflection confirms that the (001)_{pc} AgTaO₃ film is coherently strained to the substrate. Figure 2(c) shows the overlaid rocking curves of the 002 AgTaO₃ and SrTiO₃ peaks. The full width at half maximum (FWHM) for both the film (about 20 arcsec) and substrate (about 18 arcsec) are comparable and suggest high crystalline quality of the film. Figure 2(d) shows the 222 peak of 17 nm thick AgTaO₃ film grown on a SrTiO₃ (111) substrate. An RSM around the 201 substrate peak confirms that the film is coherently strained to the substrate. The FWHM for both the film (about 64 arcsec) and substrate (about 76 arcsec) are comparable.

Using the full θ -2 θ scan and following the Nelson-Riley procedure,³⁶ for the film grown on SrTiO₃(001), the out-of-plane pseudocubic lattice constant is resolved $a_{pc} = 3.9429$ Å. Our out-of-plane lattice constant is larger than the AgTaO₃ film with similar thickness grown on SrTiO₃(001) by pulsed-laser deposition (3.935 Å).²⁶ A comparable discrepancy has recently been reported between the measured and expected lattice parameters for KTaO₃ films grown on SrTiO₃, where lattice expansion was attributed to the emergence of a ferroelectric state.²⁹ The unequivocal resolution of the space group symmetries and the nature of expanded out-

of-plane lattice constant require complementary characterizations. For the $(111)_{pc}$ AgTaO_3 the extracted lattice spacing $a_{\text{ATO}_{111}}$ is 2.283 Å, which is comparable to the expected 2.285 Å of the unperturbed rhombohedral crystal structure. This close agreement is likely due to the small lattice mismatch with (111) SrTiO_3 (−0.1%).

Imaging by high-angle annular dark-field in a scanning transmission electron microscope (HAADF-STEM) was used to further investigate the AgTaO_3 films. The $(001)_{pc}$ AgTaO_3 sample shows a rather featureless image, suggesting a high crystalline quality (Fig. 3). We do not observe any extended defects and the interface appears coherent. We also used EDX elemental mapping to characterize the grown films (Fig. S6). EDX elemental maps and the corresponding intensity profiles show a familiar narrow transition region in which the silver and tantalum signals decrease while the strontium and titanium signals emerge, indicating cation intermixing across the $\text{AgTaO}_3/\text{SrTiO}_3$ interface. Although a two-dimensional electron gas can accommodate the electrostatic discontinuity at the interface,^{37,38} atomic intermixing may play a similar role.²⁸

The $(111)_{pc}$ AgTaO_3 film exhibits a high crystalline quality for the first 10 nm from the substrate interface followed by a gradually increasing disorder into a tantalum-rich oxide layer near the surface. EDX profiles display intermixing at the film and substrate interface similar to the $(001)_{pc}$ AgTaO_3 sample. The formation of the tantalum-rich disordered surface region could be associated with compensation of the highly polar (111) surface, potentially coupled with cation redistribution or silver loss during growth. EDX profiles also confirm progressive silver depletion toward the film surface, while the tantalum and oxygen signals persist (Fig. S7), consistent with the formation of the tantalum-rich surface region. Because the unreconstructed $\text{AgTaO}_3(111)_{pc}$ surface is itself polar, surface reconstruction and cation redistribution may contribute to the formation of this region. Finally, other growth-related mechanisms, including preferential silver loss, may also contribute to the tantalum-rich surface region observed.

In summary, we demonstrate MBE growth of high-quality AgTaO_3 for both $(001)_{pc}$ and $(111)_{pc}$ oriented films. XRD results suggest high crystalline perfection, and RSM confirms that the films are coherently strained to the underlying substrates. Cross-sectional HAADF-STEM confirms the high crystalline quality of the samples and does not show any extended defects. EDX results show that in both orientations, the transition region between the substrate and film is not atomically abrupt.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available within the article. **Additional data related to the film growth and structural characterization by XRD and STEM are available at <https://doi.org/10.34863/xxxx>**

REFERENCES

- ¹A. Gupta, H. Silotia, A. Kumari, M. Dumen, S. Goyal, R. Tomar, N. Wadehra, P. Ayyub, and S. Chakraverty, “KTaO₃-the new kid on the spintronics block,” *Adv. Mater.* **34**, 2106481 (2022).
- ²T. Schwaigert, A. Barooni, B. Gregory, P. Malinowski, A. Tenneti, S. Hasko, B. Palazzolo, J. W. Hodgson, B. Faeth, P. M. Woodward, K. M. Shen, A. Singer, M. Ghazisaeidi, S. Salmani-Rezaie, G. S. Schlom, and K. Ahadi, “Synthesis and Electronic Structure of the Fractionally Occupied Double Perovskite EuTa₂O₆ with Ordered Europium Vacancies,” *Adv. Funct. Mater.* **36**, e13656 (2026).
- ³A. H. Al-Tawhid, S. J. Poage, S. Salmani-Rezaie, A. Gonzalez, S. Chikara, D. A. Muller, D. P. Kumah, M. N. Gastiasoro, J. Lorenzana, and K. Ahadi, “Enhanced critical field of superconductivity at an oxide interface,” *Nano Letters* **23**, 6944–6950 (2023).

- ⁴A. H. Al-Tawhid, D. P. Kumah, and K. Ahadi, “Two-dimensional electron systems and interfacial coupling in LaCrO₃/KTaO₃ heterostructures,” *Appl. Phys. Lett.* **118**, 192905 (2021).
- ⁵K. Zou, S. Ismail-Beigi, K. Kisslinger, X. Shen, D. Su, F. Walker, and C. Ahn, “La-TiO₃/KTaO₃ interfaces: A new two-dimensional electron gas system,” *APL Materials* **3** (2015).
- ⁶C. Liu, X. Yan, D. Jin, Y. Ma, H.-W. Hsiao, Y. Lin, T. M. Bretz-Sullivan, X. Zhou, J. Pearson, B. Fisher, J. S. Jiang, W. Han, J.-M. Zuo, J. Wen, D. D. Fong, J. Sun, H. Zhou, and A. Bhattacharya, “Two-dimensional superconductivity and anisotropic transport at KTaO₃ (111) interfaces,” *Science* **371**, 716–721 (2021).
- ⁷E. G. Arnault, A. H. Al-Tawhid, S. Salmani-Rezaie, D. A. Muller, D. P. Kumah, M. S. Bahramy, G. Finkelstein, and K. Ahadi, “Anisotropic superconductivity at KTaO₃ (111) interfaces,” *Science Advances* **9**, eadf1414 (2023).
- ⁸A. H. Al-Tawhid, J. Kanter, M. Hatefipour, D. P. Kumah, J. Shabani, and K. Ahadi, “Superconductivity and weak anti-localization at ktao3 (111) interfaces: Ah al-tawhid et al.” *Journal of Electronic Materials* **51**, 6305–6309 (2022).
- ⁹A. H. Al-Tawhid, R. Sun, A. H. Comstock, D. P. Kumah, D. Sun, and K. Ahadi, “Spin-to-charge conversion at KTaO₃ (111) interfaces,” *Appl. Phys. Lett.* **126** (2025).
- ¹⁰L. M. Vicente-Arche, J. Bréhin, S. Varotto, M. Cosset-Cheneau, S. Mallik, R. Salazar, P. Noël, D. C. Vaz, F. Trier, S. Bhattacharya, A. Sander, P. Le Fèvre, F. Bertran, G. Saiz, G. Ménard, N. Bergeal, A. Barthélémy, H. Li, C.-C. Lin, D. E. Nikonov, I. A. Young, J. E. Rault, L. Vila, J.-P. Attané, and M. Bibes, “Spin–charge interconversion in KTaO₃ 2D electron gases,” *Adv. Mater.* **33**, 2102102 (2021).
- ¹¹J. Kim, M. Yu, J.-W. Lee, S.-L. Shang, G.-Y. Kim, P. Pal, J. Seo, N. Campbell, K. Eom, R. Ramachandran, M. S. Rzchowski, S. H. Oh, S.-Y. Choi, Z.-K. Liu, J. Levy, and C.-B. Eom, “Electronic-grade epitaxial (111) KTaO₃ heterostructures,” *Sci. Adv.* **10**, eadk4288 (2024).
- ¹²A. H. Al-Tawhid, J. Kanter, M. Hatefipour, D. L. Irving, D. P. Kumah, J. Shabani, and K. Ahadi, “Oxygen Vacancy-Induced Anomalous Hall Effect in a Nominally Non-magnetic Oxide,” *J. Electron. Mater.* **51**, 7073–7077 (2022).
- ¹³J. T. McCourt, E. G. Arnault, M. Baksi, S. J. Poage, S. Salmani-Rezaie, K. Ahadi, D. Kumah, and G. Finkelstein, “Electrostatic control of quantum phases in KTaO₃-based

- planar constrictions,” *Nano Letters* **25**, 16091–16096 (2025).
- ¹⁴S. J. Poage, X. Gao, M. Bakshi, S. Salmani-Rezaie, D. A. Muller, D. P. Kumah, C. N. Lau, J. Lorenzana, M. N. Gastiasoro, and K. Ahadi, “Violation of the pauli limit at $\text{ktao } 3$ (110) interfaces,” *Phys. Rev. B* **111**, 214506 (2025).
- ¹⁵L. Zhao, Q. Liu, J. Gao, S. Zhang, and J.-F. Li, “Lead-free antiferroelectric silver niobate tantalate with high energy storage performance,” *Adv. Mater.* **29**, 1701824 (2017).
- ¹⁶M. Valant, A.-K. Axelsson, and N. Alford, “Review of $\text{Ag}(\text{Nb,Ta})\text{O}_3$ as a functional material,” *J. Eur. Ceram. Soc.* **27**, 2549–2560 (2007).
- ¹⁷Z. Liu, T. Lu, J. Ye, G. Wang, X. Dong, R. Withers, and Y. Liu, “Antiferroelectrics for energy storage applications: a review,” *Adv. Mater. Technol.* **3**, 1800111 (2018).
- ¹⁸M. Francombe and B. Lewis, “Structural and electrical properties of silver niobate and silver tantalate,” *Acta Crystallogr.* **11**, 175–178 (1958).
- ¹⁹I. Belyaev, T. Lupeiko, and V. Nalbandyan, “Orthorhombic silver metatantalate and solid-solutions of system $(\text{Ag,Na})\text{TaO}_3$, $\text{Ag}(\text{Nb,Ta})\text{O}_3$,” (1978).
- ²⁰H. P. Soon, H. Taniguchi, and M. Itoh, “Dielectric and soft-mode behaviors of AgTaO_3 ,” *Phys. Rev. B* **81**, 104105 (2010).
- ²¹J. Suchanicz and A. Kania, “Uniaxial pressure effect on dielectric properties of agtao3 single crystals,” *Ferroelectrics* **393**, 21–26 (2009).
- ²²J. Li, B. G. Mullens, S. Torii, W. Ji, Z. Tan, W. Xie, T. Kamiyama, P. Miao, and B. J. Kennedy, “A high-resolution neutron diffraction and electron diffraction study of the structure and phase transitions in AgTaO_3 ,” *APL Materials* **13** (2025).
- ²³O. Diéguez, S. Tinte, A. Antons, C. Bungaro, J. B. Neaton, K. M. Rabe, and D. Vanderbilt, “Ab initio study of the phase diagram of epitaxial BaTiO_3 ,” *Phys. Rev. B* **69**, 212101 (2004).
- ²⁴N. A. Pertsev, A. G. Zembilgotov, and A. K. Tagantsev, “Effect of Mechanical Boundary Conditions on Phase Diagrams of Epitaxial Ferroelectric Thin Films,” *Phys. Rev. Lett.* **80**, 1988–1991 (1998).
- ²⁵D. G. Schlom, L.-Q. Chen, C.-B. Eom, K. M. Rabe, S. K. Streiffer, and J.-M. Triscone, “Strain tuning of ferroelectric thin films,” *Annu. Rev. Mater. Res.* **37**, 589–626 (2007).
- ²⁶T. Tachikawa, H. Y. Hwang, and Y. Hikita, “Enhancing the barrier height in oxide Schottky junctions using interface dipoles,” *Appl. Phys. Lett.* **111** (2017).

- ²⁷S. A. Khan, H. U. Khan, M. S., and Z. Ali, “Structural, electronic, optical and thermo-electric properties in the phases of AgTaO₃,” *Mater. Sci. Semicond. Process.* **122**, 105467 (2021).
- ²⁸T. Schwaigert, S. Salmani-Rezaie, M. R. Barone, H. Paik, E. Ray, M. D. Williams, D. A. Muller, D. G. Schlom, and K. Ahadi, “Molecular beam epitaxy of KTaO₃,” *J. Vac. Sci. Technol., A* **41** (2023).
- ²⁹T. Schwaigert, S. Salmani-Rezaie, S. Hazra, U. Saha, M. Ramesh, A. Ross, B. Pamuk, L.-Q. Chen, D. A. Muller, D. G. Schlom, and K. Ahadi, “Above room temperature ferroelectricity in epitaxially strained ktao3,” *Advanced Materials*, e74366 (2026).
- ³⁰K. M. Adkison, S.-L. Shang, B. J. Bocklund, D. Klimm, D. G. Schlom, and Z.-K. Liu, “Suitability of binary oxides for molecular-beam epitaxy source materials: A comprehensive thermodynamic analysis,” *APL Mater.* **8**, 081110 (2020).
- ³¹Y. A. Birkhölzer, A. S. Park, N. Schnitzer, J. Z. Kaaret, B. Z. Gregory, T. A. Kraay, T. Schwaigert, M. R. Barone, B. D. Faeth, F. V. E. Hensling, I. C. G. van den Bosch, E. M. Kiens, C. Baeumer, E. Bergamasco, M. Grüninger, A. V. Bordoallos, S. Chaulagain, N. J. Podraza, W. Tokarz, W. Tabis, M. J. Wahila, S. Sarker, C. J. Pollock, S.-L. Shang, Z.-K. Liu, N. Artrith, F. M. F. de Groot, N. A. Benedek, A. Singer, D. A. Muller, and D. G. Schlom, “Synthesis of epitaxial TaO₂ thin films on Al₂O₃ by suboxide molecular-beam epitaxy and thermal laser epitaxy,” *APL Materials* **14**, 071106 (2026).
- ³²M. Wolcyrz and M. Lukaszewski, “The crystal structure of the room-temperature phase of AgTaO₃,” *Z. Kristallogr. - Cryst. Mater.* **177**, 53–58 (1986).
- ³³T. C. Kaspar, T. Droubay, S. A. Chambers, and P. S. Bagus, “Spectroscopic evidence for Ag (III) in highly oxidized silver films by X-ray photoelectron spectroscopy,” *J. Phys. Chem. C* **114**, 21562–21571 (2010).
- ³⁴M. Khanuja, H. Sharma, B. Mehta, and S. Shivaprasad, “XPS depth-profile of the sub-oxide distribution at the native oxide/Ta interface,” *Journal of Electron Spectroscopy and Related Phenomena* **169**, 41–45 (2009).
- ³⁵W. Friedrich, P. Knipping, and M. Laue, “Interferenzerscheinungen bei roentgenstrahlen,” *Annalen der Physik* **346**, 971–988 (1913).
- ³⁶J. B. Nelson and D. P. Riley, “An experimental investigation of extrapolation methods in the derivation of accurate unit-cell dimensions of crystals,” *Proceedings of the Physical Society* **57**, 160–177 (1945).

- ³⁷K. Ahadi and S. Stemmer, “Novel metal-insulator transition at the $\text{SrTiO}_3/\text{SrTiO}_3$ interface,” *Physical Review Letters* **118**, 236803 (2017).
- ³⁸R. Mori, P. B. Marshall, K. Ahadi, J. D. Denlinger, S. Stemmer, and A. Lanzara, “Controlling a van hove singularity and fermi surface topology at a complex oxide heterostructure interface,” *Nature communications* **10**, 5534 (2019).

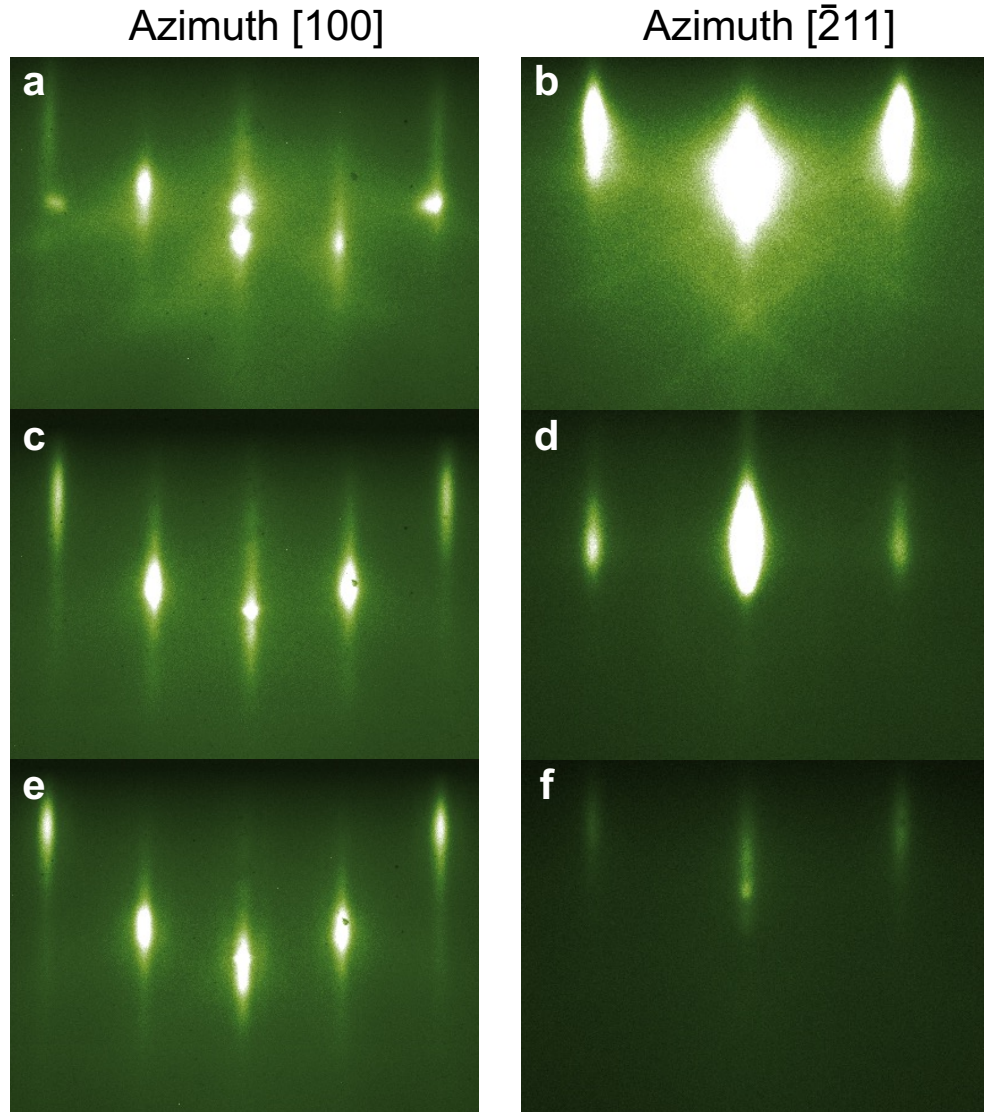


FIG. 1. RHEED pattern after growth of a formula-unit-thick layer AgTaO_3 on (a) (001) SrTiO_3 and (b) (111) SrTiO_3 substrates. RHEED pattern after growth of 20 formula-unit-thick layers of AgTaO_3 on (c) (001) SrTiO_3 and (d) (111) SrTiO_3 substrates. RHEED pattern immediately after the growth of (e) 28 nm AgTaO_3 on (001) SrTiO_3 and (f) 17 nm AgTaO_3 on (111) SrTiO_3 substrates.

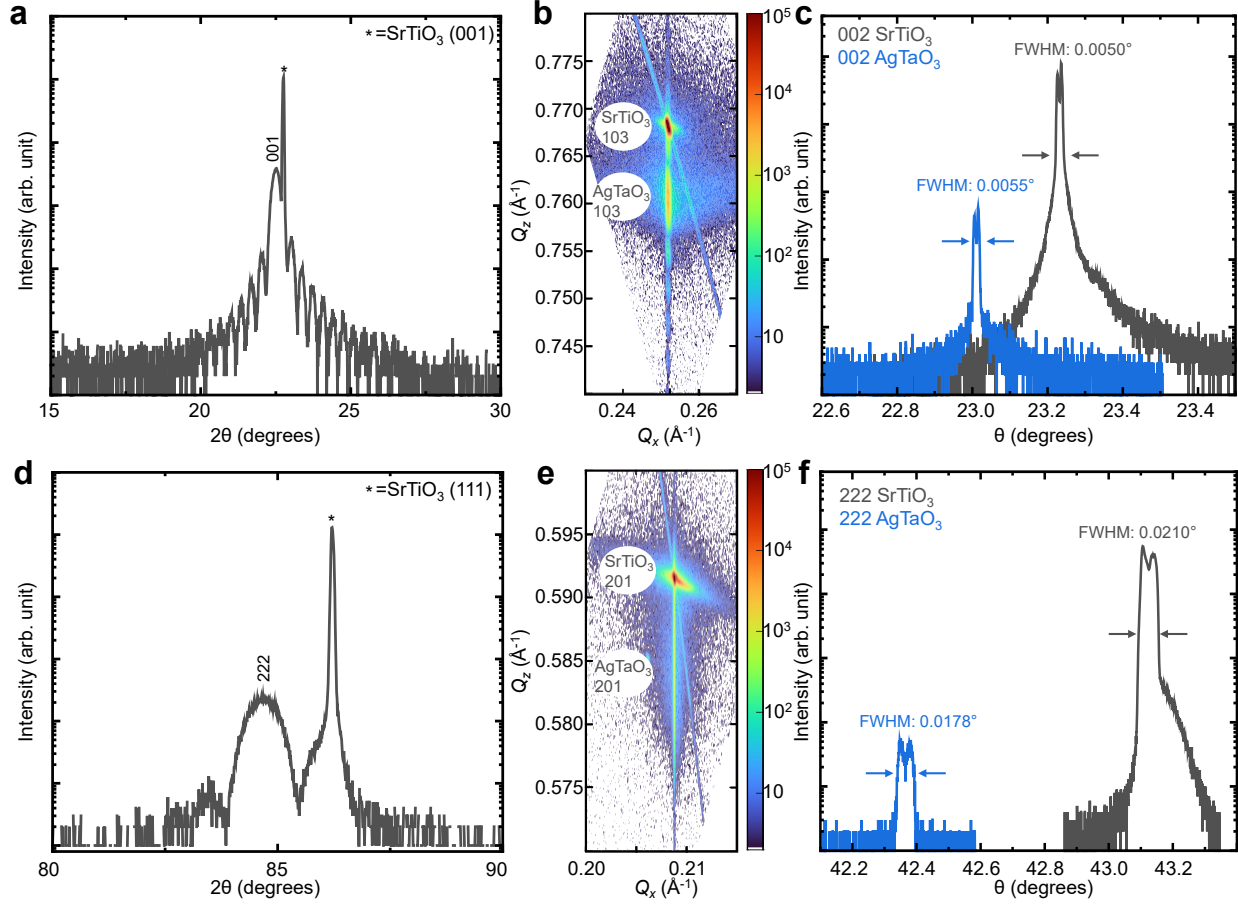


FIG. 2. (a) θ - 2θ scan, showing 001 reflection of AgTaO_3 grown on SrTiO_3 (001) substrate. Symmetric Laue fringes indicate a well-defined film thickness, indicative of an abrupt interface between film and substrate (asterisks * denote substrate reflections). (b) RSM around the 103 substrate and film reflections. The RSM results confirm that the film is fully strained to the substrate. (c) Overlaid rocking curves of the 002 SrTiO_3 and 002 AgTaO_3 reflections suggest comparable FWHMs, indicating low out-of-plane mosaicity. (d) θ - 2θ scan, showing 222 reflection of AgTaO_3 grown on a SrTiO_3 (111) substrate (asterisks * denote substrate reflections). (e) RSM around the 201 substrate reflection. The RSM results confirm that the film is coherently strained to the substrate. (f) Overlaid rocking curves of the 222 SrTiO_3 and 222 AgTaO_3 reflections exhibit comparable FWHMs, suggesting low out-of-plane mosaicity.

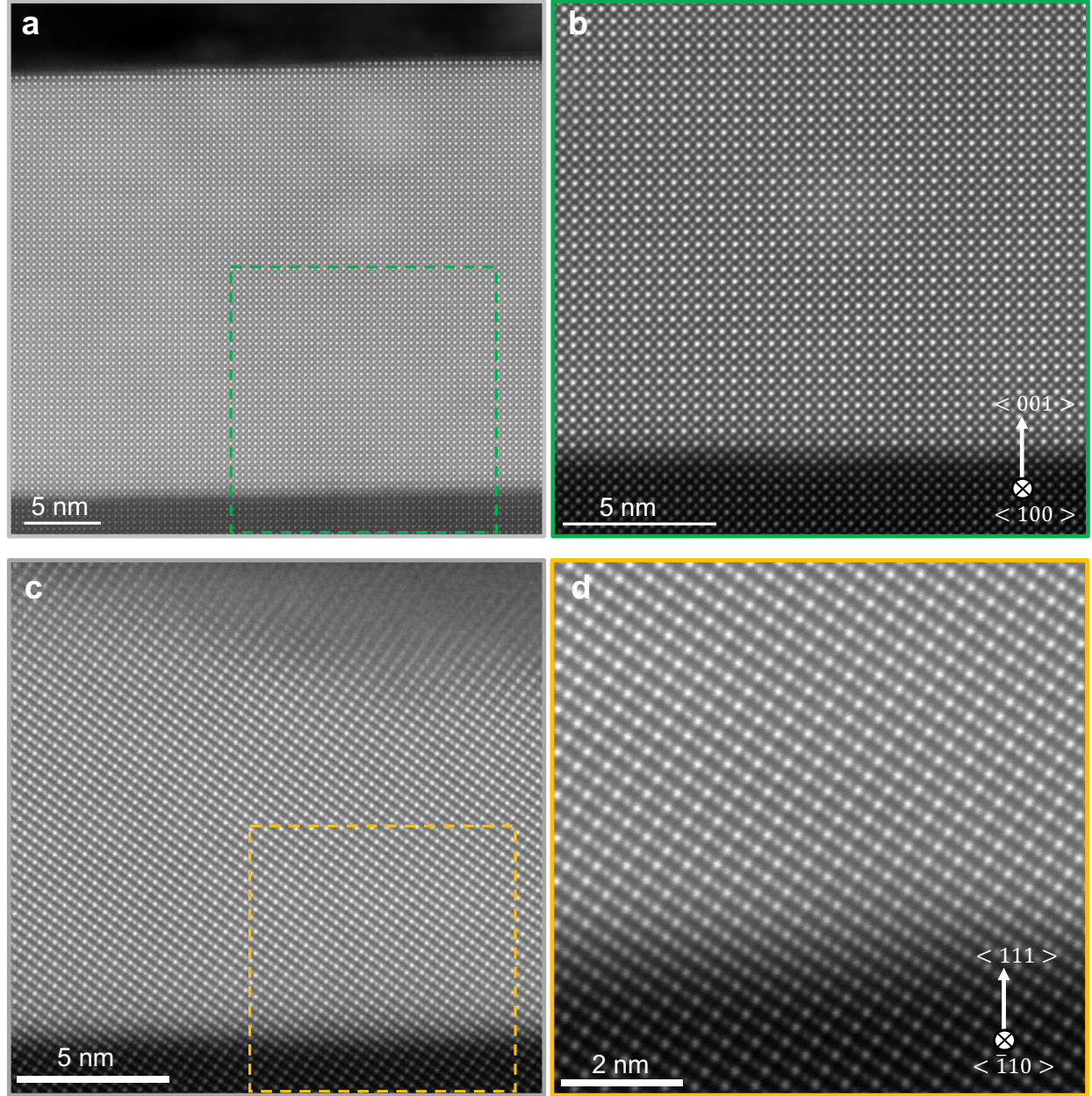


FIG. 3. Cross-section HAADF-STEM images of the same 28 nm thick (001) and 17 nm (111) AgTaO_3 films. (a) At low magnification surface steps are visible with extended regions of uniformity for the (001) AgTaO_3 film. (b) Higher magnification reveal well-resolved atomic columns within the film and a sharp interface between the film and underlying (001) SrTiO_3 substrate. (c) (111) AgTaO_3 at low magnification reveals ≈ 10 nm of well-ordered crystalline film, above which the structure transitions into a tantalum-rich oxide layer. (d) Higher magnification reveals the sharp film-substrate interface.